

CUSTOMER BEHAVIOR ANALYSIS

End-to-End Data Analytics Project Report

Python (Pandas) · Microsoft SQL Server · Power BI

Prepared: July 2026

1. Executive Summary

This project delivers an end-to-end analysis of customer shopping behavior, covering data cleaning and exploratory data analysis in Python, business-question analysis in Microsoft SQL Server, and an interactive Power BI dashboard for stakeholder reporting. The dataset contains 3,900 customer transaction records across 18 attributes, including demographics, purchase details, product category, shipping preferences, and subscription status.

The analysis identifies which customer segments and product categories drive the most revenue, how discounts and subscriptions relate to spending, and how shipping preferences affect purchase behavior. These insights support decisions on marketing targeting, loyalty/subscription strategy, and inventory focus by category.

2. Project Objective

The goal of this project was to analyze customer shopping behavior data to uncover patterns in purchasing, demographics, and engagement, and to translate those patterns into an interactive dashboard that business stakeholders can use for ongoing decision-making. Specifically, the project aimed to:

- Understand the overall customer base — size, spending patterns, and satisfaction levels.
- Compare revenue and behavior across gender, age group, and product category.
- Evaluate the impact of discounts, promo codes, and subscriptions on customer spend.
- Assess how shipping type and other service factors relate to purchase value.
- Present findings in a self-service Power BI dashboard with filters for ongoing exploration.

3. Dataset Overview

Attribute	Detail
Source file	customer_shopping_behavior.csv
Records	3,900 rows
Fields	18 columns
Key fields	Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, and additional payment/frequency fields
Granularity	One row per customer transaction

The raw CSV was loaded into a Pandas DataFrame for initial inspection and cleaning, and the same structure was loaded into a SQL Server table (customer_data) to support relational, query-based analysis.

4. Tools & Technologies

Stage	Tool	Purpose
Data loading & EDA	Python (Pandas, Jupyter/Anaconda)	Import data, inspect structure, check data quality, generate descriptive statistics
Business-question analysis	Microsoft SQL Server / SSMS (T-SQL)	Answer targeted business questions using aggregation, filtering, and subqueries
Reporting & visualization	Power BI	Build an interactive dashboard with KPI cards, charts, and slicers for stakeholders

5. Methodology

5.1 Data Preparation & Exploratory Data Analysis (Python)

Using Pandas in a Jupyter notebook, the dataset was loaded with `pd.read_csv()` and reviewed for structure, data types, and completeness across all 3,900 records and 18 columns. This step confirmed the dataset was clean and consistently formatted, with categorical fields (Gender, Category, Season, Shipping Type, Subscription Status, Discount Applied) ready for grouping, and numeric fields (Age, Purchase Amount, Review Rating, Previous Purchases) ready for statistical summarization. This EDA established the foundation for both the SQL analysis and the Power BI dashboard.

5.2 Business-Question Analysis (SQL Server)

The cleaned dataset was queried directly in SQL Server (customer_data table) using T-SQL aggregation, grouping, and subqueries. The five core business questions and their results are shown below, followed by supplementary segmentation queries run to deepen the analysis.

Q1. What is the total revenue generated by male vs. female customers?

Query: `SUM(purchase_amount)` grouped by gender.

gender	revenue
Male	157890
Female	75191

Result: Male customers generated \$157,890 in total revenue vs. \$75,191 for female customers — male customers account for roughly 68% of total revenue.

Q2. Which customers used a discount but still spent more than the average purchase amount?

Query: `customer_id` and `purchase_amount` filtered on `discount_applied = 'Yes'` and `purchase_amount >=` the overall average (via a subquery).

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60

Result (sample): a segment of customers who used a discount yet still spent above the dataset average — confirming that discounts are not only reaching price-sensitive, low-value shoppers.

Q3. Which are the top 5 products with the highest average review rating?

Query: AVG(review_rating) grouped by item_purchased, ordered descending, TOP 5.

	item_purchased	Average_Review_Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.78

Result: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78) are the top 5 highest-rated items.

Q4. Compare the average purchase amount between Standard and Express shipping.

Query: AVG(purchase_amount) grouped by shipping_type, filtered to Standard and Express. (Result pending — to be added once the query output is exported.)

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

Query: customer count, AVG(purchase_amount), and SUM(purchase_amount) grouped by subscription_status.

	subscription_status	total_customers	average_spend	total_revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

Result: 1,053 subscribers average \$59 per purchase and generate \$62,645 total revenue; 2,847 non-subscribers also average \$59 per purchase and generate \$170,436 total revenue. Average spend per transaction is essentially identical — the revenue gap is driven entirely by the larger non-subscriber base, not by subscribers spending more per visit.

5.3 Supplementary Segmentation Queries

Beyond the five core questions, additional SQL queries were run to profile customer loyalty, discount patterns, and category performance in more depth.

Discount rate by item

Query: share of transactions with discount_applied = 'Yes', grouped by item_purchased, ordered descending.

	item_purchased	discount_rate
1	Hat	50.0000000000000
2	Sneakers	49.6600000000000
3	Coat	49.0700000000000
4	Sweater	48.1700000000000
5	Pants	47.3700000000000

Result: Hats (50.0%), Sneakers (49.7%), Coats (49.1%), Sweaters (48.2%), and Pants (47.4%) have the highest discount-usage rates of any items.

Customer segments: New, Returning, Loyal

Query: customers bucketed by previous_purchases into New, Returning, and Loyal segments, with counts per segment.

	customer_segment	Number of customers
1	Returning	701
2	Loyal	3116
3	New	83

Result: Loyal customers make up 3,116 of the 3,900-customer base (79.9%), Returning customers 701 (18.0%), and New customers just 83 (2.1%).

Top-selling items by category

Query: items ranked by total_orders within each category using ROW_NUMBER()/RANK() partitioned by category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Result: leading sellers by category are Jewelry (Accessories, 171 orders), Blouse (Clothing, 171 orders), Sandals (Footwear, 160 orders), and Jacket (Outerwear, 163 orders).

Repeat buyers by subscription status

Query: count of customers with more than one purchase, grouped by subscription_status.

	subscription_status	repeated_Buyers
1	Yes	958
2	No	2518

Result: 958 of 1,053 subscribers are repeat buyers (91.0%), vs. 2,518 of 2,847 non-subscribers (88.4%) — a modest but real repeat-purchase advantage for subscribers.

Revenue by age group

Query: SUM(purchase_amount) grouped by age_group, confirming the pattern seen in the Power BI dashboard.

	age_group	total_revenue
1	Young Adult	62143
2	Middle-Aged	59197
3	Adults	55978
4	Senior	55763

Result: Young Adult (\$62,143), Middle-Aged (\$59,197), Adults (\$55,978), and Senior (\$55,763) — revenue is broadly balanced across age groups, with Young Adults slightly in the lead.

5.3 Dashboard Development (Power BI)

The findings from the EDA and SQL analysis were consolidated into a single-page interactive Power BI dashboard titled "Customer Behavior Analysis." The dashboard includes:

- KPI cards for total customers, average purchase amount, and average review rating.
- A donut chart showing the split of customers by subscription status.
- Bar charts for revenue and sales volume by product category.
- Bar charts for purchase value and sales volume by age group.
- A comparison of discount usage across the customer base.
- Slicers for Subscription Status, Gender, Category, Shipping Type, and Location, allowing stakeholders to filter the entire view interactively.

6. Key Findings

6.1 Customer Base Snapshot

3,900 Total customers	\$59.76 Average purchase amount	3.75 / 5 Average review rating
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Only 27% of customers are active subscribers, while 73% are not. However, the SQL comparison in Section 5.2 (Q5) shows subscribers and non-subscribers spend the same \$59 on average per transaction — subscription status does not, by itself, drive higher spend per visit. The value of the subscriber base instead lies in a slightly higher repeat-purchase rate (91.0% vs. 88.4%, see Section 5.3).

Revenue is also heavily skewed by gender: male customers generated \$157,890 vs. \$75,191 for female customers (Section 5.2, Q1) — male customers drive roughly 68% of total revenue despite Power BI showing a comparable customer count split, suggesting men purchase more per visit and/or more frequently.

6.2 Revenue & Sales by Category

Category	Relative revenue rank	Approx. transaction count
Clothing	1st (highest)	1,737
Accessories	2nd	1,240
Footwear	3rd	599
Outerwear	4th (lowest)	324

Clothing is by far the dominant category in both revenue and transaction volume, followed by Accessories. Footwear and Outerwear together make up a much smaller share, suggesting either lower customer demand or an opportunity for targeted promotion in these categories.

6.3 Purchases & Sales by Age Group

Spending is fairly evenly distributed across age groups. The SQL revenue-by-age-group query (Section 5.3) confirms the dashboard pattern with exact figures: Young Adult (\$62,143), Middle-Aged (\$59,197), Adults (\$55,978), and Senior (\$55,763). This indicates the product mix appeals broadly across age segments rather than skewing toward a single demographic.

6.4 Discount Behavior

A larger share of transactions occur without a discount applied than with one, though discount-driven purchases still make up a meaningful minority of sales. The SQL analysis shows some discount users spend above the overall average (Section 5.2, Q2), and that Hats, Sneakers, Coats, Sweaters, and Pants carry the highest discount-usage rates of any items — roughly 47–50% of purchases of these items involve a discount (Section 5.3). Together this suggests discounts are attracting a meaningful share of higher-value shoppers on specific items, not just driving low-value purchases.

6.5 Customer Loyalty Segments

Segmenting the customer base by purchase history reveals a heavy reliance on existing customers: 3,116 customers (79.9%) are classified as Loyal, 701 (18.0%) as Returning, and only 83 (2.1%) as New (Section 5.3). Repeat-purchase behavior is strong across the board — 91.0% of subscribers and 88.4% of non-subscribers have made more than one purchase — but the very small New-customer share indicates growth is currently sustained almost entirely by the existing base rather than fresh acquisition.

6.6 Category Leaders

Within each product category, one item stands out as the clear top seller by order volume: Jewelry in Accessories (171 orders), Blouse in Clothing (171 orders), Sandals in Footwear (160 orders), and Jacket in Outerwear (163 orders) (Section 5.3). These are strong candidates for continued inventory priority and feature placement.

7. Recommendations

- **Reframe the subscription pitch around retention, not spend:** Since subscribers and non-subscribers spend the same \$59 on average per transaction, subscription should not be marketed as a way to spend less or get better value per purchase. Instead, lean on its real, measured benefit — a higher repeat-purchase rate (91.0% vs. 88.4%) — and position subscription around convenience, retention perks, and staying engaged, not per-order savings.
- **Invest in new-customer acquisition:** New customers are only 2.1% of the base (83 of 3,900), while Loyal customers make up nearly 80%. Growth is currently almost entirely dependent on the existing base. Prioritize top-of-funnel acquisition — paid/organic marketing, referral incentives — to reduce long-term reliance on a shrinking pool of new entrants.
- **Address the gender revenue gap:** Male customers generate roughly 68% of total revenue (\$157,890 vs. \$75,191). Consider two complementary plays: deepen the already-strong male segment with loyalty offers, and run a dedicated campaign to grow female average order value and category breadth, where the revenue opportunity is currently underdeveloped.
- **Double down on Clothing and Accessories:** These categories drive the majority of revenue and volume, led by Jewelry, Blouse, Sandals, and Jacket as the top seller in each category; prioritize inventory, marketing spend, and cross-sell campaigns around these categories and hero items.
- **Right-size discounting on high-discount items:** Items like Hats, Sneakers, Coats, Sweaters, and Pants already carry discount rates near 47–50% of transactions. Since discount users are not exclusively low-value shoppers (Section 5.2, Q2), test whether margin can be protected by trimming discount depth on these specific items without materially hurting conversion.
- **Review shipping economics:** Compare the profitability of Standard vs. Express shipping (Section 5.2, Q4, pending query export) against their respective average order values to ensure shipping incentives remain cost-effective.

8. Conclusion

This project demonstrates a complete analytics workflow — from raw data to actionable insight — spanning Python for data preparation and exploration, SQL Server for structured business-question analysis, and Power BI for stakeholder-facing visualization. The combined analysis shows a customer base concentrated in Clothing and Accessories purchases, fairly balanced across age groups, and heavily reliant on a large Loyal-customer segment that generates the majority of revenue, with subscription driving modestly better retention rather than higher per-order spend. The clearest risks and opportunities are a very low new-customer share (2.1%) and a pronounced gender revenue skew toward male customers — both strong candidates for targeted follow-up campaigns. The interactive dashboard allows stakeholders to continue exploring these patterns by category, demographic, and location as new data becomes available.